hasklib: Stochastic Process Derivations

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Abstract

This document collects the stochastic-calculus results that the stochastic layer of `hasklib` relies on: from the probability-space setup, Brownian motion, Itô SDE, to Geometric Brownian Motion , Ornstein–Uhlenbeck, Arithmetic Brownian Motion, Cox–Ingersoll–Ross, and numerical schemes. Standard measure-theoretic probability is assumed; see Shreve (2004) or Williams (1991) for construction.

Contents

Chapter 1

Foundations: Itô Calculus and the SDE Framework

1.1 Probability space and Brownian motion

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. The filtration $\{\mathcal{F}_t\}$ represents the information available up to time t.

Definition 1.1 (Brownian motion)**.**
A standard Brownian motion $\{W_t\}_{t\geq 0}$ is a process with

1. $W_0 = 0$ almost surely;
2. Independent increments: for $s < t < u$, $W_u - W_t$ is independent of $W_t - W_s$;
3. Gaussian increments: $W_t - W_s \sim \mathcal{N}(0,\, t-s)$;
4. Continuous sample paths.

Remark 1.1.
In code, property (3) is what `NormalRng::draw()` supplies: a step of length Δt uses $W_{t+\Delta t} - W_t = \sqrt{\Delta t}\, Z$ with $Z \sim \mathcal{N}(0,1)$. This substitution is the single bridge between the continuous theory here and the discrete simulation code.

One might notice (2) and (4) pull in different directions. Independent increments mean each step of the path is a fresh draw, carrying no memory of the past — nothing about $\mathcal{F}_t$ constrains $W_{t+h} - W_t$. Continuity, on the other hand, means the path never jumps: W_{t+h} must stay close to W_t as $h \to 0$. It is not obvious a single process can do both at once, and the price of doing so turns out to be that the path is extremely rough and nowhere differentiable.

Fixing t and considering the difference quotient over a step $h > 0$. By the Gaussian-increment property, $W_{t+h} - W_t \sim \mathcal{N}(0, h)$, so the difference quotient

$$\frac{W_{t+h} - W_t}{h}$$

is itself Gaussian with mean 0 and variance

$$\mathrm{Var}\left(\frac{W_{t+h} - W_t}{h}\right) = \frac{1}{h^2}\,\mathrm{Var}(W_{t+h} - W_t) = \frac{h}{h^2} = \frac{1}{h},$$

using that dividing a random variable by a constant h scales its variance by $1/h^2$. Hence the standard deviation of the difference quotient is $1/\sqrt{h}$, which diverges as $h \downarrow 0$: the path moves by order $\sqrt{h}$ over a step of order h, so no finite slope survives the limit. This informal argument is made rigorous by the Paley–Wiener–Zygmund theorem (Paley, Wiener & Zygmund 1933): with probability one, $t \mapsto W_t$ is nowhere differentiable. It also follows that the path has infinite first-order (total) variation on every interval, so integrals $\int f \, \mathrm{d}W_s$ cannot be defined pathwise as Riemann–Stieltjes integrals. Its quadratic variation, however, is finite, and it is precisely this finite second-order variation that the Itô integral and Itô's lemma are built on.

1.2 Quadratic variation

The property that distinguishes stochastic from ordinary calculus is that Brownian motion accumulates quadratic variation at a nonzero rate.

Proposition 1.1 (Quadratic variation of Brownian motion).
For a standard Brownian motion, the quadratic variation over $[0, T]$ *satisfies*

$$\langle W \rangle_T \;=\; \lim_{\|\Pi\| \to 0} \sum_k \big(W_{t_{k+1}} - W_{t_k}\big)^2 \;=\; T \qquad \textit{in } L^2,$$

where Π *is a partition of* $[0, T]$*. Informally this is written*

$$(\mathrm{d}W_t)^2 = \mathrm{d}t.$$

Proof.
Write the partition as $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ and abbreviate

$$\Delta t_k = t_{k+1} - t_k, \qquad \Delta W_k = W_{t_{k+1}} - W_{t_k}, \qquad S_\Pi = \sum_{k=0}^{n-1} \big(\Delta W_k\big)^2.$$

"Convergence in L^2" means the mean-square error vanishes: $\mathbb{E}\big[(S_\Pi - T)^2\big] \to 0$ as the mesh $\|\Pi\| := \max_k \Delta t_k \to 0$. We use only two facts about the increments, both directly from the definition of Brownian motion:

- Each $\Delta W_k \sim \mathcal{N}(0, \Delta t_k)$.
- Increments over disjoint intervals are independent.

We then calculate the mean of S_Π, i.e $\mathbb{E}[S_\Pi]$
Since $\mathbb{E}[\Delta W_k] = 0$, the second moment is the variance:

$$\mathbb{E}\big[(\Delta W_k)^2\big] = \mathrm{Var}(\Delta W_k) = \Delta t_k.$$

Summing over k,

$$\mathbb{E}[S_\Pi] = \sum_k \mathbb{E}\big[(\Delta W_k)^2\big] = \sum_k \Delta t_k = T.$$

This holds for every partition, not merely in the limit. Because the mean is exactly T, the mean-square error is the variance:

$$\mathbb{E}\big[(S_\Pi - T)^2\big] = \mathrm{Var}(S_\Pi).$$

The whole statement then reduces to showing $\mathrm{Var}(S_\Pi) \to 0$.

We then calculate the variance of a single term.
Normalising the increment as $\Delta W_k = \sqrt{\Delta t_k}\, Z$ with $Z \sim \mathcal{N}(0, 1)$, so $(\Delta W_k)^2 = \Delta t_k\, Z^2$ and

$$\mathrm{Var}\big((\Delta W_k)^2\big) = (\Delta t_k)^2\ \mathrm{Var}(Z^2).$$

For a standard normal, $\mathbb{E}[Z^2] = 1$ and $\mathbb{E}[Z^4] = 3$, hence $\mathrm{Var}(Z^2) = \mathbb{E}[Z^4] - \big(\mathbb{E}[Z^2]\big)^2 = 3-1 = 2$. (The fourth moment is one Gaussian integration by parts: with the density ϕ satisfying $\phi'(z) = -z\,\phi(z)$, integrating $\int z^4 \phi \,\mathrm{d}z$ by parts gives $\mathbb{E}[Z^4] = 3\,\mathbb{E}[Z^2] = 3$.) Therefore,

$$\mathrm{Var}\left((\Delta W_k)^2\right) = 2\,(\Delta t_k)^2.$$

Step 3 Variance of the sum, then let the mesh shrink.
The increments ΔW_k are independent, meaning the squared increments $(\Delta W_k)^2$ are independent too, and the variance of a sum of independent variables is the sum of the variances:

$$\mathrm{Var}(S_\Pi) = \sum_k \mathrm{Var}\left((\Delta W_k)^2\right) = 2\sum_k (\Delta t_k)^2.$$

Bounding one factor of Δt_k in each term by the mesh:

$$\sum_k (\Delta t_k)^2 \le \Big(\max_k \Delta t_k\Big) \sum_k \Delta t_k = \|\Pi\|\, T.$$

Combining,

$$\mathbb{E}\big[(S_\Pi - T)^2\big] = \mathrm{Var}(S_\Pi) \le 2\,\|\Pi\|\, T \xrightarrow[\|\Pi\|\to 0]{} 0,$$

which is exactly the claimed L^2 convergence. □

Remark 1.2.
The mean computation ($\mathbb{E}[S_\Pi] = T$) shows the average comes out exactly right; the variance computation shows it comes out reliably, not just on average. This second part matters because a single term is not close to its own mean: $(\Delta W_k)^2$ has standard deviation $\sqrt{2}\,\Delta t_k$, about the same size as its mean Δt_k itself. The reason the full sum S_Π still ends up close to T is that the terms are independent, so their ups and downs cancel out when added together — this cancellation is what the variance computation shows, and it is a property of the sum, not of any single term. Inside a sum (equivalently, an integral), the random $(\Delta W_k)^2$ may therefore be replaced by the deterministic Δt_k: this replacement is the entire content of the shorthand $(\mathrm{d}W_t)^2 = \mathrm{d}t$ used in the next section.

1.3 The Itô stochastic differential equation

Definition 1.2 (Itô SDE).
An Itô process X_t solves the stochastic differential equation

$$\mathrm{d}X_t \;=\; a(t, X_t)\,\mathrm{d}t \;+\; b(t, X_t)\,\mathrm{d}W_t, \tag{1.1}$$

interpreted as the integral equation

$$X_t = X_0 + \int_0^t a(s, X_s)\,\mathrm{d}s + \int_0^t b(s, X_s)\,\mathrm{d}W_s,$$

where the second integral is an Itô integral. We call a the drift and b the diffusion coefficients.

Remark 1.3 (Correspondence with the code).
Equation (1.1) is exactly what the `StochasticProcess` base class encodes: `drift(t,x)` returns $a(t, x)$ and `diffusion(t,x)` returns $b(t, x)$. Every subsequent process (GBM, OU, CIR) is a choice of these two functions.

Assumption 1.1 (Existence and uniqueness).
We assume a and b satisfy the standard Lipschitz and linear-growth conditions guaranteeing a unique strong solution to (1.1); see Øksendal (2003), Thm. 5.2.1.

1.4 Itô's lemma

Itô's lemma is the chain rule for functions of an Itô process, and it is the engine behind every closed-form solution used in `hasklib` (the GBM exact sampler, the OU solution).

Theorem 1.1 (Itô's lemma, one dimension).
Let X_t solve (1.1) *and let $\varphi(t,x) \in C^{1,2}$. Then*

$$\mathrm{d}\varphi(t, X_t) = \Big(\frac{\partial\varphi}{\partial t} + a\,\frac{\partial\varphi}{\partial x} + \frac{1}{2}b^2\frac{\partial^2\varphi}{\partial x^2}\Big)\mathrm{d}t \;+\; b\,\frac{\partial\varphi}{\partial x}\,\mathrm{d}W_t, \tag{1.2}$$

Informal derivation.
This is the standard second-order Taylor argument. It is not a proof in the strict sense, as a rigorous version requires the construction of the Itô integral (Shreve 2004, Sec. 4.2.1), which is outside the scope of this document. The derivation stems around the idea that, because $\mathrm{d}X$ carries a Brownian term of size $\sqrt{\mathrm{d}t}$, a contribution that ordinary calculus discards as "second order" is in fact first order and should be kept.

Over a step of length $\mathrm{d}t$ the increment $\mathrm{d}W_t$ has typical size $\sqrt{\mathrm{d}t}$ (it is $\mathcal{N}(0, \mathrm{d}t)$).
Ranking the elementary products by their power of $\mathrm{d}t$:

$$\mathrm{d}t = O(\mathrm{d}t), \qquad \mathrm{d}W_t = O(\mathrm{d}t^{1/2}), \qquad (\mathrm{d}W_t)^2 = O(\mathrm{d}t), \qquad \mathrm{d}t\,\mathrm{d}W_t = O(\mathrm{d}t^{3/2}), \qquad (\mathrm{d}t)^2 = O(\mathrm{d}t^2).$$

Discarding everything of order smaller than $\mathrm{d}t$, and using proposition 1.1 to replace $(\mathrm{d}W_t)^2$ by its in-sum limit $\mathrm{d}t$, gives the Itô multiplication table

$$(\mathrm{d}W_t)^2 = \mathrm{d}t, \qquad \mathrm{d}t\,\mathrm{d}W_t = 0, \qquad (\mathrm{d}t)^2 = 0. \tag{1.3}$$

The last two entries are not literally zero; they are negligible next to $\mathrm{d}t$ and vanish in the mesh limit for exactly the reason the fluctuations did in section 1.2.

Taylor expansion:
Expand $\varphi(t + \mathrm{d}t,\, X_t + \mathrm{d}X)$ about (t, X_t) to second order:

$$\mathrm{d}\varphi = \frac{\partial\varphi}{\partial t}\,\mathrm{d}t + \frac{\partial\varphi}{\partial x}\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\varphi}{\partial t^2}\,(\mathrm{d}t)^2 + \frac{\partial^2\varphi}{\partial t\partial x}\,\mathrm{d}t\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\varphi}{\partial x^2}\,(\mathrm{d}X)^2 + \cdots$$

Substitute the SDE $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ and square:

$$(\mathrm{d}X)^2 = a^2(\mathrm{d}t)^2 + 2\,ab\,\mathrm{d}t\,\mathrm{d}W_t + b^2(\mathrm{d}W_t)^2.$$

Apply (1.3). In $(\mathrm{d}X)^2$ the first two terms die and only $b^2(\mathrm{d}W_t)^2 = b^2\,\mathrm{d}t$ survives:

$$(\mathrm{d}X)^2 = b^2\,\mathrm{d}t.$$

The two remaining second-order terms of the expansion vanish for the same reason:

$$\frac{1}{2}\frac{\partial^2\varphi}{\partial t^2}\,(\mathrm{d}t)^2 = 0, \qquad \frac{\partial^2\varphi}{\partial t\partial x}\,\mathrm{d}t\,\mathrm{d}X = \frac{\partial^2\varphi}{\partial t\partial x}\big(a\,(\mathrm{d}t)^2 + b\,\mathrm{d}t\,\mathrm{d}W_t\big) = 0.$$

Leaving:

$$\mathrm{d}\varphi = \frac{\partial\varphi}{\partial t}\,\mathrm{d}t + \frac{\partial\varphi}{\partial x}\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\varphi}{\partial x^2}\,b^2\,\mathrm{d}t.$$

Substitute $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ once more and collect the $\mathrm{d}t$ and $\mathrm{d}W_t$ parts:

$$\mathrm{d}\varphi = \Big(\frac{\partial\varphi}{\partial t} + a\,\frac{\partial\varphi}{\partial x} + \frac{1}{2}b^2\frac{\partial^2\varphi}{\partial x^2}\Big)\mathrm{d}t + b\,\frac{\partial\varphi}{\partial x}\,\mathrm{d}W_t,$$

which is (1.2). □

Remark 1.4.
Had X been a smooth (finite-variation) path, $(\mathrm{d}X)^2$ would have been genuinely $O((\mathrm{d}t)^2)$ and dropped out, leaving the ordinary chain rule $\mathrm{d}\varphi = \frac{\partial\varphi}{\partial t}\,\mathrm{d}t + \frac{\partial\varphi}{\partial x}\,\mathrm{d}X$. The whole gap between stochastic and ordinary calculus is the single surviving term $\frac{1}{2}b^2\frac{\partial^2\varphi}{\partial x^2}\,\mathrm{d}t$, and it survives only because $(\mathrm{d}W_t)^2 = \mathrm{d}t$ rather than 0 — i.e. because Brownian motion has nonzero quadratic variation. Every closed-form result in the code that "corrects a drift by half a variance" is exactly this one term.

Chapter 2

Geometric Brownian Motion

2.1 The GBM SDE

Definition 2.1 (Geometric Brownian motion).
X_t follows geometric Brownian motion with drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$ if it solves

$$\mathrm{d}X_t = \mu X_t \,\mathrm{d}t + \sigma X_t \,\mathrm{d}W_t, \qquad X_0 > 0. \tag{2.1}$$

Referring back to the previous chapter on Itô Calculus, this is the choice $a(t,x) = \mu x$, $b(t,x) = \sigma x$ in the general SDE $\mathrm{d}X_t = a(t, X_t)\mathrm{d}t + b(t, X_t)\mathrm{d}W_t$.

Remark 2.1 (Correspondence with the code).
This is exactly why the pair of functions `GBM::drift` and `GBM::diffusion` return: `drift(t,x)` $= \mu x$, `diffusion(t,x)` $= \sigma x$. `EulerMaruyama` consumes these polymorphically through the `StochasticProcess` interface; the sampler derived below is the exact alternative to that discretisation, used by `GBM::sample_terminal` instead of stepping the Euler scheme forward.

Both a and b are affine in x, hence globally Lipschitz and of linear growth on all of $\mathbb{R}$ (assumption 1.1), so (2.1) has a unique strong solution for any $X_0 \in \mathbb{R}$. In particular this holds for $X_0 > 0$; positivity of X_t itself is established below once the explicit solution is derived.

2.2 Linearising via Itô's lemma

The SDE (2.1) has no constant-coefficient closed form as written, because both coefficients scale with X_t. The standard fix is to apply Itô's lemma to $\varphi(t,x) = \ln x$: this removes the x-dependence from both coefficients and turns (2.1) into an SDE with constant drift and diffusion, which integrates directly.

Theorem 2.1 (Exact terminal law of GBM).
Let X_t solve (2.1)*. Then for any $T > 0$,*

$$X_T = X_0 \exp\Big(\Big(\mu - \tfrac{1}{2}\sigma^2\Big)T + \sigma W_T\Big), \tag{2.2}$$

where $W_T \sim \mathcal{N}(0,T)$.

Proof.
Apply Itô's lemma (theorem 1.1) to $\varphi(t,x) = \ln x$, so:
$\frac{\partial\varphi}{\partial t} = 0$, $\frac{\partial\varphi}{\partial x} = \frac{1}{x}$, $\frac{\partial^2\varphi}{\partial x^2} = -\frac{1}{x^2}$. With $a(t, X_t) = \mu X_t$ and $b(t, X_t) = \sigma X_t$,

$$\mathrm{d}(\ln X_t) = \Big(0 + \mu X_t \cdot \frac{1}{X_t} + \tfrac{1}{2}\sigma^2 X_t^2 \cdot \Big(-\frac{1}{X_t^2}\Big)\Big)\mathrm{d}t + \sigma X_t \cdot \frac{1}{X_t}\,\mathrm{d}W_t.$$

The X_t factors cancel completely, leaving

$$\mathrm{d}(\ln X_t) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{2.3}$$

The right-hand side no longer depends on X_t: it is Brownian motion with constant drift $\mu - \frac{1}{2}\sigma^2$ and constant volatility σ, so (2.3) integrates directly from 0 to T:

$$\ln X_T - \ln X_0 = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)T + \sigma\,(W_T - W_0) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)T + \sigma W_T,$$

using $W_0 = 0$. Exponentiating gives (2.2). Since $W_T \sim \mathcal{N}(0, T)$ by definition of Brownian motion, this is a complete description of the law of X_T given X_0. □

Remark 2.2.
Equation (2.2) is sampled with a single normal draw: generate $Z \sim \mathcal{N}(0,1)$, set $W_T = \sqrt{T}\,Z$, and evaluate the right-hand side. No time-stepping is needed, which is what makes it an exact sampler rather than a discretisation, and it is exactly what `GBM::sample_terminal` implements using `NormalRng::draw()`.

Remark 2.3 (Where the $-\frac{1}{2}\sigma^2$ comes from).
The drift that appears in (2.2) is not μ, the drift of X_t itself, but $\mu - \frac{1}{2}\sigma^2$. That correction is exactly the $\frac{1}{2}b^2\frac{\partial^2\varphi}{\partial x^2}$ term of Itô's lemma, which is the term with no analogue in ordinary calculus, mentioned in the previous chapter. If one (incorrectly) applied the ordinary chain rule to $\ln X_t$, the correction would be missing and the resulting sampler would be biased upward in expectation, as proposition 2.1 below makes precise.

2.3 Moments

The moment-matching test in `test_stochastic.cpp` checks simulated terminal samples against the following closed-form mean and variance.

Lemma 2.1 (Moment generating function of a Gaussian).
If $Y \sim \mathcal{N}(m, v)$ then $\mathbb{E}[e^Y] = \exp\big(m + \frac{1}{2}v\big)$.

Proof.
Write $Y = m + \sqrt{v}\,Z$ with $Z \sim \mathcal{N}(0,1)$. Complete the square in the Gaussian integral:

$$\mathbb{E}[e^Y] = e^m\,\mathbb{E}\big[e^{\sqrt{v}\,Z}\big] = e^m \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\, e^{\sqrt{v}\,z - \frac{1}{2}z^2}\,\mathrm{d}z = e^m\, e^{\frac{1}{2}v} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}(z-\sqrt{v})^2}\,\mathrm{d}z = e^{m+\frac{1}{2}v},$$

the last integral being 1 since it is a $\mathcal{N}(\sqrt{v}, 1)$ density integrated over $\mathbb{R}$. □

Proposition 2.1 (Mean of GBM).
$\mathbb{E}[X_T] = X_0\, e^{\mu T}$.

Proof.
By theorem 2.1, $X_T = X_0\, e^Y$ with $Y = (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T$.
Since $W_T \sim \mathcal{N}(0, T)$, $Y \sim \mathcal{N}\big((\mu - \frac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$.
Applying lemma 2.1 with $m = (\mu - \frac{1}{2}\sigma^2)T$ and $v = \sigma^2 T$,

$$\mathbb{E}[X_T] = X_0\,\mathbb{E}[e^Y] = X_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)T + \tfrac{1}{2}\sigma^2 T\Big) = X_0\, e^{\mu T}.$$

Where the $-\frac{1}{2}\sigma^2 T$ Itô correction and the $+\frac{1}{2}\sigma^2 T$ from the moment generating function cancel exactly, leaving the mean growing at the "natural" rate μ despite the correction appearing inside every individual path. □

Proposition 2.2 (Variance of GBM)**.**

$$\mathrm{Var}(X_T) = X_0^2\, e^{2\mu T}\big(e^{\sigma^2 T} - 1\big).$$

Proof.
$\mathrm{Var}(X_T) = \mathbb{E}[X_T^2] - \big(\mathbb{E}[X_T]\big)^2$.
For the second moment, write $X_T^2 = X_0^2\, e^{2Y}$ with Y as above, so $2Y \sim \mathcal{N}\big(2(\mu - \tfrac{1}{2}\sigma^2)T,\ 4\sigma^2 T\big)$.
By lemma 2.1 with $m = 2(\mu - \frac{1}{2}\sigma^2)T$ and $v = 4\sigma^2 T$,

$$\mathbb{E}[X_T^2] = X_0^2 \exp\Big(2\big(\mu - \tfrac{1}{2}\sigma^2\big)T + 2\sigma^2 T\Big) = X_0^2\, e^{(2\mu+\sigma^2)T}.$$

Subtracting $\big(\mathbb{E}[X_T]\big)^2 = X_0^2\, e^{2\mu T}$ from proposition 2.1 and factoring,

$$\mathrm{Var}(X_T) = X_0^2\, e^{(2\mu+\sigma^2)T} - X_0^2\, e^{2\mu T} = X_0^2\, e^{2\mu T}\big(e^{\sigma^2 T} - 1\big). \qquad \square$$

Remark 2.4 (What the test checks).
In the code, `test_stochastic.cpp` draws a large number of terminal samples from `GBM::sample_terminal`, computes their sample mean and sample variance, and checks these against propositions 2.1 and 2.2 to within Monte Carlo tolerance. This is a check on the sampler, not on (2.1) itself — it would catch, for instance, a sign error in the $-\frac{1}{2}\sigma^2$ correction, since that error changes the mean but leaves the qualitative shape of the distribution intact.

Chapter 3

The Ornstein–Uhlenbeck Process

Chapter 4

Arithmetic Brownian Motion

Chapter 5

The Cox–Ingersoll–Ross Process

Chapter 6

Numerical Schemes: Euler–Maruyama and Milstein

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